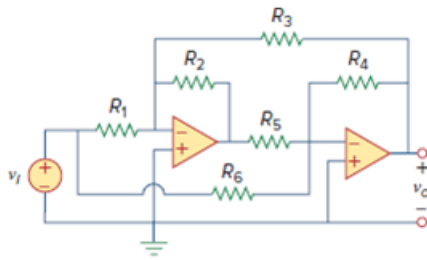


Problem

Determine the gain v_o/v_i of the circuit in Fig. 5.90.

Figure. 5.90:



Step-by-step solution

Step 1 of 3

Refer to figure 5.90 in the textbook.

As the voltage at the non-inverting terminals of both the operational amplifiers is 0 V, the voltage at the inverting terminals of the both the operational amplifiers is 0 V. Since, both the terminals of operational amplifier are virtually grounded.

Redraw the circuit as shown in figure 1.

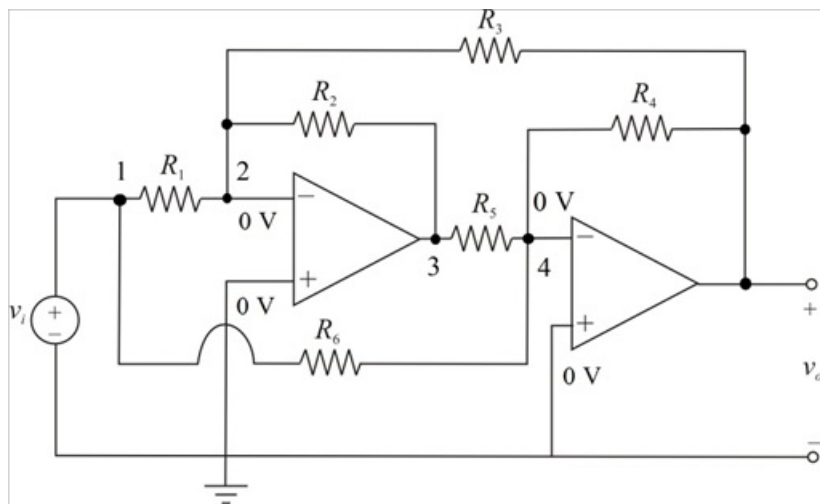


Figure 1

Step 2 of 3

Assume that voltage at node 3 is V_1 .

Apply Kirchhoff's current law at node 2 of the circuit shown in figure 1.

$$\frac{0-v_i}{R_1} + \frac{0-v_o}{R_3} + \frac{0-V_1}{R_2} = 0 \dots\dots (1)$$

Apply Kirchhoff's current law at node 4 of the circuit shown in figure 1.

$$\frac{0-V_1}{R_5} + \frac{0-v_o}{R_4} + \frac{0-v_i}{R_6} = 0$$

$$V_1 = -R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right)$$

Step 3 of 3

Substitute $-R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right)$ for V_1 in equation (1).

$$\frac{0-v_i}{R_1} + \frac{0-v_o}{R_3} + \frac{0 - \left[-R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right) \right]}{R_2} = 0$$

$$\frac{0-v_i}{R_1} + \frac{0-v_o}{R_3} + \frac{R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right)}{R_2} = 0$$

$$v_o \left(\frac{R_5}{R_2 R_4} - \frac{1}{R_3} \right) = v_i \left(\frac{1}{R_1} - \frac{R_5}{R_2 R_6} \right)$$

$$\frac{v_o}{v_i} = \frac{\frac{1}{R_1} - \frac{R_5}{R_2 R_6}}{\frac{R_5}{R_2 R_4} - \frac{1}{R_3}}$$

Simplify further,

$$\frac{v_o}{v_i} = \frac{\left(\frac{R_2 R_6 - R_1 R_5}{R_1 R_2 R_6} \right)}{\left(\frac{R_3 R_5 - R_2 R_4}{R_2 R_3 R_4} \right)}$$

$$\frac{v_o}{v_i} = \frac{R_3 R_4}{R_1 R_6} \left(\frac{R_2 R_6 - R_1 R_5}{R_3 R_5 - R_2 R_4} \right)$$

$$\frac{v_o}{v_i} = \frac{\frac{R_2 R_3 R_4}{R_1} - \frac{R_5 R_4 R_5}{R_6}}{R_3 R_5 - R_2 R_4}$$

Simplify further,

$$\frac{v_o}{v_i} = \frac{R_3 R_5 \left(\frac{R_2 R_4}{R_1 R_5} - \frac{R_4}{R_6} \right)}{R_3 R_5 - R_2 R_4}$$

$$\frac{v_o}{v_i} = \frac{\frac{R_2 R_4}{R_1} - \frac{R_4}{R_6}}{1 - \frac{R_2 R_4}{R_3 R_5}}$$

Therefore, the gain $\frac{v_o}{v_i}$ is $\boxed{\frac{\frac{R_2 R_4}{R_1} - \frac{R_4}{R_6}}{1 - \frac{R_2 R_4}{R_3 R_5}}}$.